CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.4 Electromagnetic Spectrum

Syllabus Section: 7. Waves **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (63 Marks)

Question 1 Easy [8 marks]

Part (a) [2 marks]

EM waves consist of oscillations in two perpendicular fields.

State the names of the fields.

Mark Scheme (2 marks):

The perpendicular fields in which EM waves oscillate are:

- Electric; *[1 mark]*
- Magnetic; *[1 mark]*

[Total: 2 marks]

Part (b) [3 marks]

Table 1.1 lists the names of the waves in the EM spectrum in order of increasing frequency.

Table 1.1

EM wave
Radio
Infrared
Ultraviolet
Gamma

Complete Table 1.1 by adding in the missing information.

Mark Scheme (3 marks):

A table scoring 3 marks:

EM wave
Radio
***Microwave**; [1 mark]*
Infrared
***Visible light**; [1 mark]*
Ultraviolet
***X-rays**; [1 mark]*
Gamma

[Total: 3 marks]

Part (c) [2 marks]

State two uses for infrared radiation.

Mark Scheme (2 marks):

Uses of infrared radiation:

*Any **two** from:*

- Remote controls; *[1 mark]*
- Fibre optics; *[1 mark]*
- Wired and wireless communications; *[1 mark]*
- Thermal imaging; *[1 mark]*
- Night vision; *[1 mark]*
- Heating; *[1 mark]*
- Cooking; *[1 mark]*
- Motion sensors; *[1 mark]*

[Total: 2 marks]

You should know this already from IGCSE, but make sure you can name a couple of uses for each EM wave.

Part (d) [1 mark]

Give a value for the speed of gamma radiation travelling through a vacuum.

Mark Scheme (1 marks):

The speed of gamma radiation travelling through a vacuum is:

- 3.00×10^{8} m s^{-1} *[1 mark]*

[Total: 1 mark]

This question can be asked in many ways, but it will always have the same answer! ***All*** *EM waves travel at the speed of light!*

Question 2 Easy [12 marks]

Part (a) [3 marks]

Give the name of electromagnetic wave with the

(i) Lowest energy.

[1]

(ii) Shortest wavelength.

[1]

(iii) Lowest frequency.

[1]

Mark Scheme (3 marks):

(i) The EM wave with the lowest energy is...

- Radio; *[1 mark]*

(ii) The EM wave with the shortest wavelength is...

- Gamma; *[1 mark]*

(iii) The EM wave with the lowest frequency is...

- Radio; *[1 mark]*

[Total: 3 marks]

The shorter the wavelength, the higher the frequency. The higher the frequency, the higher the energy. You need to be super confident in these relationships. Questions may be asked in terms of increasing or decreasing wavelength and frequency. Make sure you are never caught out by this.

Part (b) [2 marks]

State the range of wavelengths for infrared radiation.

Mark Scheme (2 marks):

EM radiation falls between...

- 1×10^{-3} m *[1 mark]*
- 7×10^{-7} m *[1 mark]*

[Total: 2 marks]

You do need to know the range of wavelengths for the different EM waves.

Part (c) [4 marks]

Determine the range of frequencies for infrared radiation.

Mark Scheme (4 marks):

To determine the range of frequencies for infrared radiation:

List the known quantities:

- Longest wavelength = 1×10^{-3} m (from part (b))
- Shortest wavelength = 7×10^{-7} m (from part (b))
- Speed of light in free space, c = 3.00×10^{8} m s^{-1} (from the data page)

To determine the highest frequency:

- $v = f\lambda \rightarrow f = \frac{v}{\lambda}$
- $f_H = \frac{3.00\times10^{8}}{7\times10^{-7}}$ *[1 mark]*
- $f_H = 4\times10^{14}$ Hz *[1 mark]*

To determine the lowest frequency:

- $f_L = \frac{3.00\times10^{8}}{1\times10^{-3}}$ *[1 mark]*
- $f_L = 3\times10^{11}$ Hz *[1 mark]*

[Total: 4 marks]

Remember that to calculate the highest frequency you need to use the shortest wavelength and vice versa.

The expected final answers here are expressed to 1 significant figure, consistent with the data given. Marks are awarded for correct working and a correct final answer — you will not be penalised for expressing a correct answer to more significant figures

Part (d) [3 marks]

The visible region of the electromagnetic spectrum has frequencies ranging from 4.3×10^{14} Hz to 7.5×10^{14} Hz.

Show that the wavelength of visible light is 400 nm to 700 nm.

Mark Scheme (3 marks):

To show that the wavelength of visible light is 400 nm to 700 nm:

List the known quantities:

- Lowest frequency = 4.3×10^{14} Hz
- Highest frequency = 7.5×10^{14} Hz
- Speed of light in free space, c = 3.00×10^{8} m s^{-1} (from the data page)

To determine the longest wavelength:

- $v = f\lambda \rightarrow \lambda = \frac{v}{f}$ *[1 mark]*
- $\lambda_L = \frac{3.00 \times 10^8}{4.3 \times 10^{14}} = 7.0 \times 10^{-7}$ m $= 700 \times 10^{-9}$ m $= 700$ nm *[1 mark]*

To determine the shortest wavelength:

- $\lambda_S = \frac{3.00 \times 10^8}{7.5 \times 10^{14}} = 4 \times 10^{-7}$ m $= 400 \times 10^{-9}$ m $= 400$ nm *[1 mark]*

[Total: 3 marks]

Remember that to calculate the longest wavelength you need to use the lowest frequency and vice versa.

Question 3 Medium [4 marks]

Part (a) [1 mark]

List the electromagnetic waves in order of increasing wavelength.

Mark Scheme (1 marks):

The electromagnetic waves in order of increasing wavelength are:

- Gamma
- X-rays
- Ultraviolet
- Visible light
- Infrared
- Microwaves
- Radio waves

All correct for 1 mark

[Total: 1 mark]

Part (b) [2 marks]

State the two types of electromagnetic radiation that can knock electrons out of orbit.

Mark Scheme (2 marks):

Two types of EM radiation that can cause ionisation are:

*Any **two** from:*

- Gamma; *[1 mark]*
- X-rays; *[1 mark]*
- Ultraviolet; *[1 mark]*

[Total: 2 marks]

Part (c) [1 mark]

Complete the following sentence about electromagnetic radiation by choosing the correct word.

wavelength	**frequency**	**amplitude^2**

> The energy of an electromagnetic wave is directly proportional to .....
>
> **Mark Scheme (1 marks):**
>
> The energy of an electromagnetic wave is directly proportional to....
>
> - Frequency; *[1 mark]*
>
> ***[Total: 1 mark]***
>
> *Energy is **inversely proportional** to wavelength (the shorter wavelengths have the highest energy), and **intensity** is **proportional** to amplitude squared.*

Question 4 Medium [12 marks]

Part (a) [3 marks]

Complete the table by stating the correct electromagnetic wave for each range of wavelengths.

EM wave	Range of wavelengths / m
	10^{-1} - 10^{6}
	4×10^{-7} - 7×10^{-7}
	10^{-16} - 10^{-10}

Mark Scheme (3 marks):

A table scoring three marks:

EM wave	Range of wavelengths / m
***Radio waves**; [1 mark]*	10^{-1} - 10^{6}
***Visible light**; [1 mark]*	4×10^{-7} - 7×10^{-7}
***Gamma**; [1 mark]*	10^{-16} - 10^{-10}

[Total: 3 marks]

Part (b) [6 marks]

Determine the range of frequencies for the three waves given in part (a).

Mark Scheme (6 marks):

To determine the range of frequencies for the three waves given in part (a):

Radio waves:

- $\lambda_{R\ max} = 1\times10^{6}$ m
- $\lambda_{R\ min} = 1\times10^{-1}$ m

- Speed of light in free space, $c = 3.00 \times 10^{8}$ m s^{-1}
- Using the wave speed equation:
  - $v = f\lambda \Rightarrow f = \frac{c}{\lambda}$
  - $f_{R\ min} = \frac{3.00 \times 10^{8}}{1 \times 10^{6}} = 300$ Hz *[1 mark]*
  - $f_{R\ max} = \frac{3.00 \times 10^{8}}{1 \times 10^{-1}} = 3.00 \times 10^{9}$ Hz *[1 mark]*

Visible light:

- $\lambda_{V\ max} = 7 \times 10^{-7}$ m
- $\lambda_{V\ min} = 4 \times 10^{-7}$ m
- Using the wave speed equation:
  - $f_{V\ min} = \frac{3.00 \times 10^{8}}{7 \times 10^{-7}} = 4.3 \times 10^{14}$ Hz *[1 mark]*
  - $f_{V\ max} = \frac{3.00 \times 10^{8}}{4 \times 10^{-7}} = 7.5 \times 10^{14}$ Hz *[1 mark]*

Gamma radiation:

- $\lambda_{G\ max} = 1 \times 10^{-10}$ m
- $\lambda_{G\ min} = 1 \times 10^{-16}$ m
- Using the wave speed equation:
  - $f_{G\ min} = \frac{3.00 \times 10^{8}}{1 \times 10^{-10}} = 3 \times 10^{18}$ Hz *[1 mark]*
  - $f_{G\ max} = \frac{3.00 \times 10^{8}}{1 \times 10^{-16}} = 3 \times 10^{24}$ Hz *[1 mark]*

[Total: 6 marks]

*Remember that to find the **minimum** frequency of a range you need to use the **maximum** wavelength and vice versa.*

Part (c) **[3 marks]**

Suggest a reasonable estimate for the wavelength of green light. Explain your answer.

Mark Scheme (3 marks):

A reasonable estimate for the wavelength of green light is...

- 550 nm **OR** 5.5 × 10^{-7} m *[1 mark]*

Explanation:

- There are seven colours in the visible spectrum and green is exactly in the middle; *[1 mark]*
  - Red, orange, yellow, green, blue, indigo, violet

- The range of visible light is 400 nm - 700 nm, so green light is exactly in the middle of this range; *[1 mark]*
    - Range = $700 \quad - \quad 400 \quad = \quad 300 \quad \text{nm}$
    - Half of range = $\frac{300}{2} \quad = \quad 150 \quad \text{nm}$
    - Min wavelength + half of range $= 400 \quad + \quad 150 \quad = \quad 550 \quad \text{nm}$

[Total: 3 marks]

Question 5 Medium [9 marks]

Part (a) [1 mark]

State the three types of EM radiation that are used for communications.

Mark Scheme (1 marks):

The three types of EM radiation that are used for communications are...

- Radio waves
- Microwaves
- Infrared

All correct for 1 mark

[Total: 1 mark]

Part (b) [4 marks]

State the properties common to all electromagnetic radiation.

Mark Scheme (4 marks):

The properties common to all types of EM radiation are:

- Travel at the speed of light (3.00×10^8 m s^{-1}) in free space; *[1 mark]*
- Transverse **OR** oscillations are perpendicular to the direction of energy transfer; *[1 mark]*
- Progressive waves **OR** carry energy; *[1 mark]*
- Can be polarised / diffracted / refracted / reflected / undergo interference; *[1 mark]*

[Total: 4 marks]

Part (c) [4 marks]

Dogs can see wavelengths in the range of 430 - 475 nm which they perceive as blue, and 500 - 620 nm which they perceive as yellow.

Determine the range of frequencies which dog's eyes can detect.

Mark Scheme (4 marks):

To determine the range of frequencies which dog's eyes can detect:

List the known quantities:

Blue:

- $\lambda_{B\ max} = 475\ \text{nm} = 475 \times 10^{-9}\ \text{m}$
- $\lambda_{B\ min} = 430\ \text{nm} = 430 \times 10^{-9}\ \text{m}$

Yellow:

- $\lambda_{Y\ max} = 620\ \text{nm} = 620 \times 10^{-9}\ \text{m}$
- $\lambda_{Y\ min} = 500\ \text{nm} = 500 \times 10^{-9}\ \text{m}$

From the data page:

- Speed of light in free space, $c = 3.00 \times 10^{8}\ \text{m}\ \text{s}^{-1}$

To determine the range for (perceived) blue light:

- Using the wave speed equation, $v = f\lambda \quad \Rightarrow \quad f = \frac{c}{\lambda}$
  - $f_{B\ max} = \frac{3.00 \times 10^{8}}{430 \times 10^{-9}} = 7.0 \times 10^{14}\ \text{Hz}$ *[1 mark]*
  - $f_{B\ min} = \frac{3.00 \times 10^{8}}{475 \times 10^{-9}} = 6.32 \times 10^{14}\ \text{Hz}$ *[1 mark]*

To determine the range for (perceived) yellow light:

- $f_{Y\ max} = \frac{3.00 \times 10^{8}}{500 \times 10^{-9}} = 6 \times 10^{14}\ \text{Hz}$ *[1 mark]*
- $f_{Y\ min} = \frac{3.00 \times 10^{8}}{620 \times 10^{-9}} = 4.8 \times 10^{15}\ \text{Hz}$ *[1 mark]*

[Total: 4 marks]

Remember that when calculating ranges, to find the maximum frequency you need to use the minimum wavelength and vice versa.

Question 6 Hard [7 marks]

Part (a) [2 marks]

Justify the approximation of the speed of light in air as 3.00×10^{8} m s^{-1}.

Mark Scheme (2 marks):

Justify the approximation of the speed of light in air as 3.00×10^{8} m s^{-1}:

- 3.00×10^{8} m s^{-1} is the speed of light in free space / a vacuum; *[1 mark]*
- The reduction in speed as light travels through air is negligible (therefore 3.00×10^{8} m s^{-1} is a reasonable approximation); *[1 mark]*

[Total: 2 marks]

*The CIE IAL Physics specification defines **justify** as 'support a case with evidence / argument'.*

Therefore, you must provide evidence (speed of light in free space) and then present your argument or reason as to why this is a reasonable approximation of the speed of light in air (that the decrease in speed is negligible).

Part (b) [2 marks]

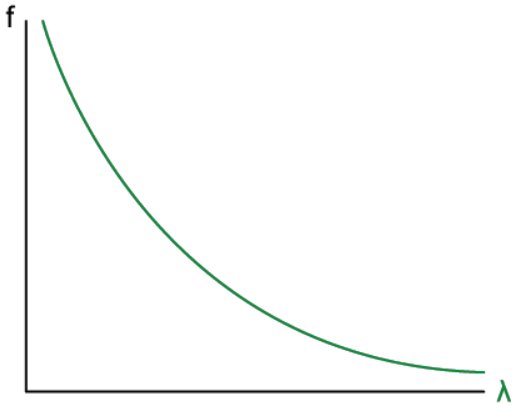
Frequency and wavelength are fundamental properties of all waves. The relationship between them depends on the context of interaction and on the type of wave in question.

Sketch a graph showing the relationship between the frequency and wavelength of an electromagnetic wave.

Mark Scheme (2 marks):

The relationship between the frequency and wavelength of an electromagnetic wave:

- A sketch graph that would score 2 marks:



- Inverse proportion curve; *[1 mark]*
- Axes labelled frequency / *f* **AND** wavelength / *λ*; *[1 mark]*

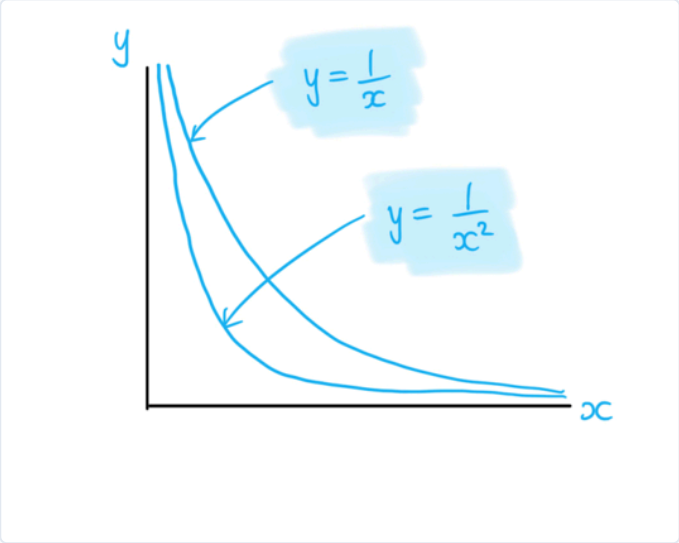
[Total: 2 marks]

The axes could be labelled either way around because the relationship is still inversely proportional.

This question assesses your understanding of the fact that the speed of electromagnetic waves is constant. We know the equation for wave speed is given by:

$$v = f\lambda$$

Therefore, if the speed v is constant, then it is true that f is inversely proportional to λ. Examiners sometimes like to test your recognition of inverse-proportionality graphs – not to be confused with inverse-square relationships (which look steeper). This distinction is sketched below:



Part (c) [3 marks]

Electromagnetic waves carry energy through the vacuum of space as progressive transverse waves.

Describe the motion of an electromagnetic wave with reference to the appropriate fields.

Mark Scheme (3 marks):

Describe the motion of the fields which comprise an electromagnetic wave:

- Electromagnetic waves are made of electric and magnetic fields; *[1 mark]*
- These fields oscillate at 90 degrees / perpendicular to each other; *[1 mark]*

Describe the direction of energy transfer:

- (Because electromagnetic waves are transverse,) energy transfer is 90 degrees / perpendicular to the direction of (field) oscillations; *[1 mark]*

[Total: 3 marks]

Question 7 Hard [11 marks]

Part (a) [2 marks]

Microwaves can be used in communications between satellites.

Calculate the frequency range for microwaves.

Mark Scheme (2 marks):

To determine the range of frequencies for microwave radiation:

List the known quantities:

- Speed of light in free space, c = 3.00×10^{8} m s^{-1} (from the data page)
- Range of wavelengths for microwave radiation = 0.1 m - 1×10^{-3} m *[1 mark]*

Determine the highest and lowest frequency:

- $v = f\lambda \rightarrow f = \frac{v}{\lambda}$
- $f_H = \frac{3.00\times10^{8}}{1\times10^{-3}} = 3\times10^{11}$ Hz
- $f_L = \frac{3.00\times10^{8}}{0.1} = 3\times10^{9}$ Hz
- Range of frequencies of microwave radiation = 3×10^{11} Hz - 3×10^{9} Hz *[1 mark]*

[Total: 2 marks]

You are expected to be able to recall the ranges of wavelengths for the different waves in the EM spectrum.

Part (b) [3 marks]

A satellite passing Neptune communicates with Earth using a microwave transmitter with an output power of 24.0 W and wavelength 78 900 μm.

The satellite's controller is located on Earth, at a distance of 4.40×10^{12} m when the signal is transmitted.

For this communication

(i) Calculate the time taken for the signal to be detected by the controller.

[1]

(ii) Calculate the energy of a microwave photon.

[2]

Mark Scheme (3 marks):

(i) Calculate the time taken for the signal to be detected by the controller:

List the known quantities:

- Distance from satellite to controller, s = 4.40 × 10^{12} m
- Speed of light in free space, c = 3.00 × 10^{8} m s^{-1} (from the data page)

Using the speed equation, calculate time taken:

- $v = \frac{d}{t} \rightarrow t = \frac{d}{v}$
- $t = \frac{4.40 \times 10^{12}}{3.00 \times 10^{8}}$ = 1.47 × 10^{4} s *[1 mark]*

(ii) Calculate the energy of a microwave photon:

List the known quantities:

- Wavelength of microwave, λ = 78 900 μm
- Planck constant, h = 6.63 × 10^{-34} J s (from the data page)

Combine the photon energy and wave speed equations to give a formula for energy using wavelength:

- Wave speed, $v = f\lambda$ and photon energy, $E = hf$
- $E = \frac{hc}{\lambda}$ *[1 mark]*

Substitute the known values to calculate the energy of a microwave photon:

- $E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{78900 \times 10^{-6}}$ = 2.52 × 10^{-24} J (3 s.f.) *[1 mark]*

[Total: 3 marks]

Part (c) **[6 marks]**

The controller dish aerial has an effective area of 258 m^2.

For the communication from part (b)

(i) Determine the power received by the controller dish aerial. You may assume that the power transmitted by the satellite radiates uniformly in all directions.

[2]

(ii) The actual power received by the controller dish aerial is 1.4 × 10^{-15} W. Suggest why this is different to the calculated power received.

[2]

(iii) Calculate the rate at which microwave photons arrive at the controller dish aerial.

[2]

Mark Scheme (6 marks):

(i) Determine the power received by the controller dish aerial:

List the known quantities:

- Output power of satellite = 24.0 W (from part (b))
- Distance from satellite to controller = 4.40×10^{12} m
- Area of controller dish aerial = 258 m^2

Devise a formula for the amount of power reaching the controller dish:

- Surface area of a sphere, $A = 4\pi r^2$
- Since the power transmitted by the satellite is assumed to radiate uniformly in all directions
- $P = \frac{\text{Output power}}{4\pi r^2} \times \text{area of controller dish}$ *[1 mark]*

Substitute the known values to calculate the power reaching the controller dish:

- $P = \frac{24.0}{4\pi \times 4.40 \times 10^{12}{}^2} \times 258 = 2.55 \times 10^{-23}$ W (3 s.f.) *[1 mark]*

(ii) The actual power received by the controller dish aerial is 1.4×10^{-15} W. Suggest why this is different to the calculated power received:

- The calculation assumed that power is radiated uniformly in all directions, but (in reality) the signal emitted by the satellite is directed to Earth (by a dish); *[1 mark]*
- Therefore, the calculated power received is less than the actual power received because less power is lost (radiating in different directions); *[1 mark]*

(iii) Calculate the rate at which the microwave photons actually arrive at the controller dish aerial:

List the known quantities:

- Energy of a microwave photon, $E = 2.52 \times 10^{-24}$ J (from part (b))

Devise a formula for the rate of photon detection:

- $\text{Rate of photon detection} = \frac{\text{Power received}}{\text{Energy of a microwave photon}}$ *[1 mark]*

Substitute in the known values to calculate the rate of photon detection:

- $\text{Rate of photon detection} = \frac{1.4 \times 10^{-15}}{2.52 \times 10^{-24}} = 5.6 \times 10^{8}\ s^{-1}$ (2 s.f.) *[1 mark]*

[Total: 6 marks]

This question is synoptic; it draws on content from multiple topics, as exam questions frequently do. If you need some help with the maths, it is covered in more detail in the Luminosity & Radiant Flux Revision Note.